

Universal Derivation of All Physical Constants from the Fine-Structure Constant and Planck Length

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Abstract

This document shows how the physical constants treated here can be calculated in SI units from two experimental base quantities together with the SI definitions (and, for atomic quantities, the electron mass) - the dimensionless fine-structure constant $\alpha = 1/137.036$ and the Planck length $\ell_P = 1.616255 \times 10^{-35}$ m. Additionally, the value of the characteristic energy E_0 in T0 theory is discussed: the geometric mean of the CODATA electron and muon masses gives $E_0 = 7.3479$ MeV; the value $E_0 = 7.398$ MeV used for α is interpreted here as the fractally corrected geometric mean $\sqrt{m_e m_\mu / K_{\text{frak}}}$. Two common circularity objections are discussed. The calculation reduces the number of natural constants that must be entered independently to two experimental values plus SI conventions, and the T0 raw values reproduce mass ratios without correction to about 0.5 %.

Note on versions. The German version of this document is more extensive than this English one. Earlier revisions shortened or changed parts of the English version. Where the two differ, the more extensive German version applies.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

.1 Introduction and Basic Principle

The Minimal Principle of Physics

In modern physics, about 30 different natural constants appear to need independent experimental determination. This work shows that, together with the SI definitions, the constants treated here can be calculated from **two experimental values**:

Fundamental Input Data

- **Fine-structure constant:** $\alpha = \frac{1}{137.035999084}$ (dimensionless)
- **Planck length:** $\ell_P = 1.616255 \times 10^{-35}$ m

SI Base Definitions

Additionally, we use the modern SI base definitions (since 2019):

$$\mu_0 = 1.25663706127(20) \times 10^{-6} \text{ N/A}^2 \quad (\text{measured, CODATA 2022;}) \quad (1)$$

exactly $4\pi \times 10^{-7}$ H/m until 2019)

$$e = 1.602176634 \times 10^{-19} \text{ C} \quad (\text{exact definition}) \quad (2)$$

$$k_B = 1.380649 \times 10^{-23} \text{ J/K} \quad (\text{exact definition}) \quad (3)$$

$$N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1} \quad (\text{exact definition}) \quad (4)$$

.2 Derivation of Fundamental Constants

Speed of Light c

The speed of light follows from the relationship between Planck units. Since the Planck length is defined as:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \quad (5)$$

and all Planck units are interconnected through $\hbar$, G and c , $c = \ell_P/t_P$ is the ratio of length and time in the time-mass duality (Doc. 101, 145), in natural units $c = 1$. Expressed in SI:

Speed of Light

$$c = 2.99792458 \times 10^8 \text{ m/s} \quad (6)$$

The number 299 792 458 m/s is the SI definition of the metre (1983) and serves as the conversion anchor; it is not the result of the dimensional analysis but expresses c in SI, see the section on the SI definitions.

Vacuum Permittivity ϵ_0

From the Maxwell relation $\mu_0 \epsilon_0 = 1/c^2$ follows:

$$\epsilon_0 = \frac{1}{\mu_0 c^2} \approx \frac{1}{4\pi \times 10^{-7} \times (2.99792458 \times 10^8)^2} \quad (7)$$

Vacuum Permittivity

$$\epsilon_0 = 8.8541878128 \times 10^{-12} \text{ F/m} \quad (8)$$

Since SI 2019, μ_0 is no longer exactly $4\pi \times 10^{-7}$ H/m but measured: $\mu_0 = 2\alpha\hbar/(e^2c) = 1.25663706127(20) \times 10^{-6} \text{ N/A}^2$ (CODATA 2022); exact are $\hbar$, e , c , k_B , N_A . The SI order is therefore

reversed: $\varepsilon_0 = e^2/(4\pi\hbar c\alpha)$ from the measured α , then $\mu_0 = 1/(\varepsilon_0 c^2)$, as in the section on the SI definitions. The number $8.8541878128 \times 10^{-12}$ F/m (CODATA 2018) differs from the $4\pi \times 10^{-7}$ value only by $\sim 10^{-10}$ relative.

Reduced Planck Constant $\hbar$

The fine-structure constant is defined as:

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \quad (9)$$

Solving for $\hbar$:

$$\hbar = \frac{e^2}{4\pi\varepsilon_0 c \alpha} \quad (10)$$

Substituting known values:

$$\hbar = \frac{(1.602176634 \times 10^{-19})^2}{4\pi \times 8.8541878128 \times 10^{-12} \times 2.99792458 \times 10^8 \times \frac{1}{137.035999084}} \quad (11)$$

Reduced Planck Constant

$$\hbar = 1.054571817 \times 10^{-34} \text{ J}\cdot\text{s} \quad (12)$$

Gravitational Constant G

From the definition of the Planck length follows:

$$G = \frac{\ell_P^2 c^3}{\hbar} \quad (13)$$

Substituting calculated values:

$$G = \frac{(1.616255 \times 10^{-35})^2 \times (2.99792458 \times 10^8)^3}{1.054571817 \times 10^{-34}} \quad (14)$$

Gravitational Constant

$$G = 6.67430 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2) \quad (15)$$

.3 Complete Planck Units

With $\hbar$, c and G , all Planck units can be calculated:

Planck Time

$$t_P = \sqrt{\frac{\hbar G}{c^5}} = \frac{\ell_P}{c} = 5.391247 \times 10^{-44} \text{ s} \quad (16)$$

Planck Mass

$$m_P = \sqrt{\frac{\hbar c}{G}} = 2.176434 \times 10^{-8} \text{ kg} \quad (17)$$

Planck Energy

$$E_P = m_P c^2 = \sqrt{\frac{\hbar c^5}{G}} = 1.956082 \times 10^9 \text{ J} = 1.220890 \times 10^{19} \text{ GeV} \quad (18)$$

Planck Temperature

$$T_P = \frac{E_P}{k_B} = \frac{m_P c^2}{k_B} = 1.416784 \times 10^{32} \text{ K} \quad (19)$$

.4 Atomic and Molecular Constants**Classical Electron Radius**

With the electron mass $m_e = 9.1093837015 \times 10^{-31} \text{ kg}$:

$$r_e = \frac{e^2}{4\pi\epsilon_0 m_e c^2} = \frac{\alpha \hbar}{m_e c} = 2.817940 \times 10^{-15} \text{ m} \quad (20)$$

Compton Wavelength of the Electron

$$\lambda_{C,e} = \frac{h}{m_e c} = \frac{2\pi \hbar}{m_e c} = 2.426310 \times 10^{-12} \text{ m} \quad (21)$$

Bohr Radius

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} = \frac{\hbar}{m_e c \alpha} = 5.291772 \times 10^{-11} \text{ m} \quad (22)$$

Rydberg Constant

$$R_\infty = \frac{\alpha^2 m_e c}{2h} = \frac{\alpha^2 m_e c}{4\pi \hbar} = 1.097373 \times 10^7 \text{ m}^{-1} \quad (23)$$

.5 Thermodynamic Constants**Stefan-Boltzmann Constant**

$$\sigma = \frac{2\pi^5 k_B^4}{15h^3 c^2} = \frac{2\pi^5 k_B^4}{15(2\pi \hbar)^3 c^2} = 5.670374419 \times 10^{-8} \text{ W/(m}^2 \cdot \text{K}^4) \quad (24)$$

Wien's Displacement Law Constant

$$b = \frac{hc}{k_B} \times \frac{1}{4.965114231} = 2.897771955 \times 10^{-3} \text{ m} \cdot \text{K} \quad (25)$$

.6 Dimensional Analysis and Verification

Consistency Check of the Fine-Structure Constant

$$[\alpha] = \frac{[e^2]}{[\epsilon_0][\hbar][c]} \quad (26)$$

$$= \frac{[C^2]}{[F/m][J \cdot s][m/s]} \quad (27)$$

$$= \frac{[C^2]}{[C^2 \cdot s^2 / (kg \cdot m^3)][J \cdot s][m/s]} \quad (28)$$

$$= \frac{[C^2]}{[C^2 / (kg \cdot m^2 / s^2)]} \quad (29)$$

$$= [1] \quad \checkmark \quad (30)$$

Consistency Check of the Gravitational Constant

$$[G] = \frac{[\ell_p^2][c^3]}{[\hbar]} \quad (31)$$

$$= \frac{[m^2][m^3/s^3]}{[J \cdot s]} \quad (32)$$

$$= \frac{[m^5/s^3]}{[kg \cdot m^2 / s^2 \cdot s]} \quad (33)$$

$$= \frac{[m^5/s^3]}{[kg \cdot m^2 / s^3]} \quad (34)$$

$$= [m^3 / (kg \cdot s^2)] \quad \checkmark \quad (35)$$

Consistency Check of $\hbar$

$$[\hbar] = \frac{[e^2]}{[\epsilon_0][c][\alpha]} \quad (36)$$

$$= \frac{[C^2]}{[F/m][m/s][1]} \quad (37)$$

$$= \frac{[C^2]}{[C^2 \cdot s^2 / (kg \cdot m^3)][m/s]} \quad (38)$$

$$= \frac{[C^2 \cdot kg \cdot m^3]}{[C^2 \cdot s \cdot m]} \quad (39)$$

$$= [kg \cdot m^2 / s] = [J \cdot s] \quad \checkmark \quad (40)$$

.7 The Characteristic Energy E_0 and T0 Theory

Definition of the Characteristic Energy

Basic Definition

The fundamental definition of the characteristic energy is:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (41)$$

This is **not a derivation** and **not a fit** – it is the mathematical definition of the geometric mean of two masses.

Numerical Evaluation with Different Precision Levels

Level 1: Rounded Standard Values

With the often cited rounded masses:

$$m_e = 0.511 \text{ MeV} \quad (42)$$

$$m_\mu = 105.658 \text{ MeV} \quad (43)$$

$$E_0^{(1)} = \sqrt{0.511 \times 105.658} = \sqrt{53.99} = 7.348 \text{ MeV} \quad (44)$$

Level 2: CODATA 2018 Precision Values

With the precise experimental masses:

$$m_e = 0.510,998,946,1 \text{ MeV} \quad (45)$$

$$m_\mu = 105.658,374,5 \text{ MeV} \quad (46)$$

$$E_0^{(2)} = \sqrt{0.5109989461 \times 105.6583745} = 7.347,878 \text{ MeV} \quad (47)$$

Level 3: The Value $E_0 = 7.398 \text{ MeV}$

Critical Question

Is $E_0 = 7.398 \text{ MeV}$ a fitted parameter?

Answer in this document: It is not treated as a free fit parameter.

$E_0 = 7.398 \text{ MeV}$ is the fractally corrected geometric mean $\sqrt{m_e m_\mu / K_{\text{frak}}} = 7.3479 / \sqrt{0.98667} = 7.397 \text{ MeV}$ (Doc. 190, R72); the geometric mean of the CODATA masses themselves is 7.3479 MeV (Level 2).

Precise Fine-Structure Constant Calculation

The dimensionally correct formula:

$$\alpha = \xi \cdot \frac{E_0^2}{(1 \text{ MeV})^2} \quad (48)$$

where:

- $\xi = \frac{4}{3} \times 10^{-4} = 1.333\bar{3} \times 10^{-4}$ (exact)
- $(1 \text{ MeV})^2$ is the normalization energy for dimensionless calculation

Comparison of Calculation Accuracy

E_0 Value	Source	α_{T0}^{-1}	Deviation
7.348 MeV	Rounded masses	138.91	1.37%
7.347,878 MeV	CODATA exact	138.91	1.37%
7.398 MeV	Fractally corrected	137.035	0.0005%
Experiment (CODATA):		137.035999084	Reference

Table 1: Comparison of calculation accuracy for different E_0 values

Detailed Calculation with $E_0 = 7.398$ MeV

$$E_0^2 = (7.398)^2 = 54.7303 \text{ MeV}^2 \quad (49)$$

$$\frac{E_0^2}{(1 \text{ MeV})^2} = 54.7303 \quad (50)$$

$$\alpha = 1.333\bar{3} \times 10^{-4} \times 54.7303 \quad (51)$$

$$= 7.297 \times 10^{-3} \quad (52)$$

$$\alpha^{-1} = 137.035 \quad (53)$$

Comparison with Experiment

T0 value: $\alpha^{-1} = 137.035$

Experiment: $\alpha^{-1} = 137.035999084$

Relative Deviation: $\frac{|137.0353-137.0360|}{137.036} = 0.0005\%$

Assessment: The agreement is a property of the anchor E_0 , which provides the connection to the SI; it is not counted as an independent prediction of α .

.8 On the Precision of the Value $E_0 = 7.398$ MeV

Why $E_0 = 7.398$ MeV Works Well

According to the interpretation of this document, the value $E_0 = 7.398$ MeV is meant to result from:

1. **Geometric mean calculation** of the electron and muon masses with full precision
2. **Fractal correction:** $7.398/7.348 = 1.0068 = 1/\sqrt{K_{\text{frak}}}$, i.e. $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}}$; the difference from the bare mean is not a radiative correction (QED, weak interaction)
3. **Consistency** with T0 geometry $\xi = \frac{4}{3} \times 10^{-4}$

The Mathematical Justification

Geometric Interpretation

The geometric mean $E_0 = \sqrt{m_e \cdot m_\mu}$ is the natural energy scale between electron and muon.

On a logarithmic scale, E_0 lies precisely in the middle:

$$\log(E_0) = \frac{\log(m_e) + \log(m_\mu)}{2} \quad (54)$$

This is the **characteristic energy** of the first two lepton generations.

.9 Comparison with Alternative Approaches

Estimation with T0-Calculated Masses

If the particle masses themselves were calculated from T0 theory:

$$m_e^{T0} = 0.511,000 \text{ MeV} \quad (\text{theoretical}) \quad (55)$$

$$m_\mu^{T0} = 105.658,000 \text{ MeV} \quad (\text{theoretical}) \quad (56)$$

$$E_0^{T0} = \sqrt{0.511000 \times 105.658000} = 7.348 \text{ MeV} \quad (57)$$

Result: This calculation gives $E_0 = 7.348 \text{ MeV}$, the bare value, as in Level 1.

Correct Interpretation

The correct approach is:

1. Use **experimental masses** as input
2. Calculate **geometric mean** exactly
3. Use **T0 geometry** ξ as theoretical parameter
4. Check **fine-structure constant** as output

.10 Dimensional Consistency of the E_0 Formula

Correct Dimensionless Formulation

The formula:

$$\alpha = \xi \cdot \frac{E_0^2}{(1 \text{ MeV})^2} \quad (58)$$

is dimensionally consistent:

$$[\alpha] = [\xi] \cdot \frac{[E_0^2]}{[(1 \text{ MeV})^2]} \quad (59)$$

$$= [1] \cdot \frac{[\text{Energy}^2]}{[\text{Energy}^2]} \quad (60)$$

$$= [1] \quad \checkmark \quad (61)$$

Alternative Notation

Equivalently can be written:

$$\frac{1}{\alpha} = \frac{(1 \text{ MeV})^2}{\xi \cdot E_0^2} = \frac{1}{\xi \cdot 54.73} = \frac{1}{1.333 \times 10^{-4} \times 54.73} = 137.036 \quad (62)$$

.11 On the Circularity Objections

The Apparent Circularity Objections

Common Criticisms

Objection 1: The Planck length ℓ_P is already defined via the gravitational constant G :

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \quad (63)$$

Therefore, it is circular to derive G from ℓ_P .

Objection 2: The speed of light c is calculated from μ_0 and ε_0 :

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \quad (64)$$

But ε_0 is calculated from c - that is circular.

Resolution of the Apparent Circularity

The Structure of SI Definitions (since 2019)

Modern SI Base

Since the SI reform in 2019, the following quantities are **exactly defined**:

$$c = 299792458 \text{ m/s} \quad (\text{exact definition}) \quad (65)$$

$$e = 1.602176634 \times 10^{-19} \text{ C} \quad (\text{exact definition}) \quad (66)$$

$$\hbar = 1.054571817 \times 10^{-34} \text{ J}\cdot\text{s} \quad (\text{exact definition}) \quad (67)$$

$$k_B = 1.380649 \times 10^{-23} \text{ J/K} \quad (\text{exact definition}) \quad (68)$$

μ_0 and ε_0 are no longer fixed exactly but follow from the measured α : $\mu_0 = 2\alpha\hbar/(e^2c) = 1.25663706127(20) \times 10^{-6} \text{ N/A}^2$ (CODATA 2022)

Corrected Hierarchy with Modern SI

The actual derivation is therefore:

$$\text{Given (experimental): } \alpha, \ell_P \quad (69)$$

$$\text{Defined (SI 2019): } c, e, \hbar, k_B \quad (70)$$

$$\text{Calculated: } \epsilon_0 = \frac{e^2}{4\pi\hbar c \alpha} \quad (71)$$

$$\mu_0 = \frac{1}{\epsilon_0 c^2} \quad (72)$$

$$G = \frac{\ell_P^2 c^3}{\hbar} \quad (73)$$

Result: For c , ϵ_0 and μ_0 there is no circularity, since c and $\hbar$ are directly defined. For G it should be noted that, in practice, ℓ_P has so far been determined from the measured G .

ℓ_P is Only ONE Possible Length Scale

The Planck length is not the only fundamental length scale. One could equally well use:

$$L_1 = 2.5 \times 10^{-35} \text{ m} \quad (\text{arbitrarily chosen}) \quad (74)$$

$$L_2 = 1.0 \times 10^{-35} \text{ m} \quad (\text{round number}) \quad (75)$$

$$L_3 = \pi \times 10^{-35} \text{ m} \quad (\text{with } \pi) \quad (76)$$

$$L_4 = e \times 10^{-35} \text{ m} \quad (\text{with } e) \quad (77)$$

The Mathematics Works with ANY Length Scale

The general formula is:

$$G = \frac{L^2 \times c^3}{\hbar} \quad (78)$$

Crucial: Only with the specific length $\ell_P = 1.616255 \times 10^{-35} \text{ m}$ does one obtain the correct experimental value of G .

The SI Reference is What Matters

Length Scale L	Calculated G	Status
$2.5 \times 10^{-35} \text{ m}$	$1.60 \times 10^{-10} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$	Wrong
$1.0 \times 10^{-35} \text{ m}$	$2.55 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$	Wrong
$\pi \times 10^{-35} \text{ m}$	$2.52 \times 10^{-10} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$	Wrong
$\ell_P = 1.616 \times 10^{-35} \text{ m}$	$6.674 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$	Correct

Table 2: G-values for different length scales

The Hierarchy in the T0 Interpretation

Interpretation in the T0 Framework

In the T0 interpretation, ℓ_P and G are both expressions of the same underlying geometry.

The order in this interpretation:

1. Fundamental 3D space geometry $\rightarrow \xi = \frac{4}{3} \times 10^{-4}$
2. From this follows ℓ_P as natural scale
3. From this follows G as emergent property
4. SI units provide the reference to human measures

Possible Independent Approaches

Independent Measurement of ℓ_P

The Planck length could in principle become accessible independently of G through **[S]** (no such measurement exists so far):

1. **Quantum gravity experiments:** Direct measurement of the minimal length scale
2. **Black hole Hawking radiation:** ℓ_P determines the evaporation rate
3. **Cosmological observations:** ℓ_P influences quantum fluctuations of inflation (cosmic sector, deliberately set aside in the current corpus, P39)
4. **High-energy scattering experiments:** At Planck energies, ℓ_P becomes directly accessible

Independent Measurement of α

The fine-structure constant is measured through:

1. **Quantum Hall effect:** $\alpha = \frac{e^2}{h} \times \frac{R_K}{Z_0}$
 2. **Anomalous magnetic moment:** α from QED corrections
 3. **Atom interferometry:** α from recoil measurements
 4. **Spectroscopy:** α from hydrogen spectrum
- None of these methods uses G or ℓ_P .

Formal Check of the Order of Definitions

Definition Hierarchy

$$\text{Given: } \alpha \text{ (experimental), } \ell_P \text{ (experimental)} \quad (79)$$

$$\text{Defined (SI 2019): } c, e, h \quad (80)$$

$$\text{Calculated: } \epsilon_0 = \frac{e^2}{4\pi\hbar c\alpha}, \quad \mu_0 = \frac{1}{\epsilon_0 c^2} \quad (81)$$

$$G = \frac{\ell_P^2 c^3}{\hbar} \quad (82)$$

Under SI 2019, μ_0 is no longer an SI convention but measured; the order corresponds to the corrected hierarchy above.

Each quantity depends only on previously defined quantities.

Circularity Test

A circular argument exists if:

$$A \xrightarrow{\text{defined}} B \xrightarrow{\text{defined}} C \xrightarrow{\text{defined}} A \quad (83)$$

In our case:

$$\alpha, \ell_P \xrightarrow{\text{calculated}} \varepsilon_0, \mu_0, G \nrightarrow \alpha, \ell_P \quad (84)$$

Result: Within this chain of calculation there is no circularity.

The Philosophical Argument

Reference Scales are Necessary

Reference Scales

All physics needs reference scales.

Nature is dimensionally structured. To get from dimensionless relationships to measurable quantities, we need:

- An **energy scale** (from α)
- A **length scale** (from ℓ_P)
- **SI conventions** (human measures)

This holds for any dimensional physics, not only for T0 theory.

.12 Further Considerations

Connection to the T0 Model

Within the T0 model, α and ℓ_P could also be derived from geometric principles **[S]**:

$$\xi = \frac{4}{3} \times 10^{-4} \quad (3D \text{ space geometry}) \quad (85)$$

$$\alpha = \xi \times E_0^2 \quad \text{with } E_0 = \sqrt{m_e \times m_\mu / K_{\text{frak}}} \quad (86)$$

$$\ell_P = \xi \times \ell_{\text{fundamental}} \quad (87)$$

This would reduce the number of fundamental parameters to **one**, the geometric parameter ξ (plus motivated structural choices and SI anchors for unit conversion).